

Summary

Computational Applied Mathematician with a Ph.D. in Mathematics from Claremont Graduate University and over 8 years of industrial experience in software development, data analytics, and signal processing. Skilled in designing computational solutions for science and engineering problems and delivering consulting services for advanced analytics and machine learning projects.

Education

Claremont Graduate University <i>Ph.D. Mathematics</i> Specialization in Computational and Applied Mathematics	Claremont, CA 2025
Claremont Graduate University <i>M.S. Mathematics</i>	Claremont, CA 2019
Pitzer College (Harvey Mudd College coursework) <i>B.A. Mathematics & Computer Science</i>	Claremont, CA 2017

Professional Experience

ACERTAS Analytics <i>Senior Software Engineer</i> - Develops big data solutions integrating R, Python, and C/C++ for high-performance analytics. - Applies advanced signal processing and machine learning methods to client projects in engineering and business domains. - Designs natural language processing and predictive modeling systems to support decision-making. - Collaborates directly with clients to scope, implement, and validate analytic solutions.	Claremont, CA 2017–Present
American Express (Harvey Mudd College Clinic) <i>Machine Learning Researcher (Project)</i> - Developed and enhanced a hotel recommendation system using machine learning on American Express closed-loop customer data. - Improved top-1 recommendation accuracy by over 6% through feature engineering and feature weighting approaches. - Designed and tested k-Nearest Neighbors (k-NN) models, feature construction (e.g., Millennialness, Familyness, Loyaltyness), and deep learning methods with TensorFlow. - Delivered actionable data insights on customer travel patterns, seasonal behavior, and city-specific performance to inform recommender strategy.	Claremont, CA 2016–2017
Claremont McKenna College <i>Research Assistant</i> - Conducted applied mathematics research in MATLAB and Python for image processing. - Reviewed and implemented algorithms from academic research papers.	Claremont, CA 2016
Claremont McKenna College <i>Teaching Assistant</i> Graded assignments and held tutoring sessions for courses in Data Structures and Advanced Programming.	Claremont, CA 2015–2016
Pitzer College <i>Tutor</i> Provided mathematics tutoring for Calculus I, II, and III students.	Claremont, CA 2014

Technical Skills

Interests: Machine Learning, Natural Language Processing, Signal Processing, Software Development
Programming Languages: Angular, Python, R, MATLAB, C/C++, Java, Scala, Haskell, Standard ML
Tools: TensorFlow, Git, SVN, SQLite, MongoDB, Vim, Microsoft Office, $\text{\LaTeX}$

Publications

For full list, see Google Scholar: <https://scholar.google.com/citations?user=DyFbVikAAAAJ>